

Eczema/ dermatitis

1. The terms eczema and dermatitis are synonymous.
2. Distinctive reaction patterns in the skin cause either acute or chronic eczema.

The acute stage eczema : In the acute stage, oedema of the epidermis (spongiosis) progresses to the formation of intra-epidermal vesicles, which may enlarge and rupture.

The chronic stage eczema : In the chronic stage there is less oedema and vesiculation, but more thickening of the epidermis (acanthosis); this is accompanied by a variable degree of vasodilatation and T-cell lymphocytic infiltration in the upper dermis.

Classification of eczema

- Atopic
- Seborrhoeic
- Discoid
- Irritant
- Allergic
- Asteatotic
- Gravitational
- Lichen simplex
- Pompholyx

Clinical features

1. There are several patterns of eczema; some have identifiable environmental causes, whereas others are more complex. The clinical signs are similar in all types and vary according to the duration of the rash.
2. The features of acute and chronic eczema are as follows:

The eczema reaction**Acute**

- Redness and swelling, usually with ill-defined margins
- Papules, vesicles and, more rarely, large blisters
- Exudation and cracking
- Scaling

Chronic

- May show all of the above, but usually less vesicular and exudative
- Lichenification, a dry leathery thickening with increased skin markings, secondary to rubbing and scratching
- Fissures and scratch marks
- Hypo- or hyperpigmentation

Atopic eczema

1. Atopy is a genetic predisposition to form excessive IgE, which leads to a generalised and prolonged hypersensitivity to common environmental antigens such as pollen and house dust mite.
2. Atopic individuals manifest one or more of a group of diseases that includes asthma, hay fever, food and other allergies, and atopic eczema.

Diagnostic criteria for atopic eczema

Itchy skin and at least three of the following:

- History of itch in skin creases (or cheeks if < 4 yrs)
- History of asthma/hay fever (or in a first-degree relative if < 4 yrs)
- Dry skin (xeroderma)
- Visible flexural eczema (cheeks, forehead, outer limbs if < 4 yrs)
- Onset in first 2 yrs of life

Atopic eczema: distribution and character of rash**Infancy**

- Often acute and involves the face and trunk
- Frequent sparing of the napkin area-Childhood
- Backs of the knees, fronts of the elbows, wrists and ankles

Adults

- Face and trunk are involved; lichenification is common

Complications of atopic eczema : Superinfection

Bacterial : (Staphylococcus aureus) most common

Viral

- Herpes simplex virus causes a widespread severe eruption-eczema herpeticum
- Papillomavirus and molluscum contagiosum may be encouraged by use of local corticosteroids

Irritant reactions

- Defective barrier function

Sleep disturbance

- Loss of schooling and behavioural difficulties

Food allergy

- Eggs, cow's milk, protein, fish, wheat and soya may cause an immediate urticarial eruption rather than exacerbation of eczema

Seborrhoeic eczema : This is characterised by a red scaly rash and classically affects the scalp (dandruff), central face, nasolabial folds, eyebrows and central chest. It is due to *Pityrosporum ovale* infection of the skin. When severe, it may resemble psoriasis. Seborrhoeic eczema is a feature of AIDS, in which case it can be very severe.

Discoïd eczema : This common form of eczema is recognised from discrete coin-shaped eczematous lesions, particularly on the limbs of young and elderly men. It can occur in children with atopic eczema and may be difficult to treat.

Irritant eczema

1. Strong irritants elicit an acute reaction at the site of contact, whereas weak irritants most often cause chronic eczema, especially of the hands, after prolonged exposure.
2. Detergents, alkalis, acids, solvents and abrasive dusts are common causes.
3. There is a wide range of susceptibility to weak irritants.
4. Irritant eczema accounts for the majority of occupational cases and work loss.
5. The elderly, those with fair and dry skin, and those with an atopic background are especially vulnerable.

Allergic contact eczema

1. This is due to a delayed hypersensitivity reaction following contact with antigens or haptens
2. Previous exposure to the allergen is required for sensitisation and the reaction is specific to the allergen or closely related chemicals.

Common allergens

Allergen	Present in
Nickel	Jewellery, jean studs, bra clips
Dichromate	Cement, leather, matches
Rubber chemicals	Clothing, shoes, tyres
Colophony	Sticking plaster, collodion
Paraphenylenediamine	Hair dye, clothing
Balsam of Peru	Perfumes, citrus fruits
Neomycin, benzocaine	Topical applications
Parabens	Preservative in cosmetics and creams
Wool	Lanolin, cosmetics, creams
Epoxy resin	Resin adhesives
Allergic contact eczema	

This was caused by the application of an antihistamine cream. The acute eczematous reaction and bilateral periorbital oedema are typical.

Gravitational (stasis) eczema : This occurs on the lower legs and is often associated with signs of venous insufficiency: oedema, red or bluish discoloration, loss of hair, induration, haemosiderin pigmentation and ulceration.

Lichen simplex

- This describes a plaque of lichenified eczema due to repeated rubbing or scratching, as a habit or in response to stress
- Common sites include the nape of the neck, the lower legs and the anogenital area.

Pompholyx

- Recurrent vesicles and bullae occur on the palms, palmar surface of the fingers and soles, and are excruciatingly itchy.

- This form of eczema can occur in atopic eczema and in irritant and contact allergic dermatitis. It can be provoked by heat, stress and nickel ingestion in a nickel-sensitive patient but is often idiopathic.

Seborrhoeic eczema: Antipityrosporal agents, such as ketoconazole shampoo and creams, form the basis of treatment, supplemented with weak corticosteroids if needed. Treatments may need to be repeated at intervals.

Irritant eczema: This is treated by the regular use of emollients, protective clothing (e.g. gloves) and avoidance of irritants.

Contact allergic eczema

1. Avoidance of the culprit allergen is the most important intervention; this may entail lifestyle modifications, such as a change in occupation or hobbies.
2. Measures used for irritant eczema are also helpful.

Gravitational eczema

1. Topical corticosteroids should only be applied to eczematous areas and should be avoided on ulcers.
2. Sensitisation to topical antibiotics (neomycin) and preservatives (e.g. chlorocresol) is common. Associated peripheral oedema is treated by elevation of the leg and graded compression bandages.

Investigation of eczema

1. Patch tests are performed in suspected cases of contact allergic dermatitis
2. Prick tests or IgE and specific IgE tests are occasionally performed to support the diagnosis of atopic eczema and to determine specific environmental allergens, e.g. pet dander, horse hair, house dust mite, pollens and foods.
3. Bacterial and viral swabs for microscopy and culture are useful in suspected secondary infection.
4. Bacteria are invariably present, but antibacterial treatment should be reserved for those cases

with evidence of clinical infection.

5. Individuals with atopic eczema have an increased susceptibility to herpes simplex virus (HSV), and are at risk of developing a widespread infection, eczema herpeticum. The presence of small punched-out lesions on a background of worsening eczema suggests the possibility of secondary HSV infection.

Urticaria

1. Urticaria refers to an area of focal dermal oedema secondary to a transient increase in capillary permeability.
2. On certain body sites such as the lips or hands, the oedema spreads deeper and is referred to as angioedema.
3. By definition, the swelling lasts less than 24 hours.
4. Acute urticaria may be associated with angioedema of the lips, face, throat and rarely wheezing, abdominal pain, headaches and even anaphylaxis.
5. Whilst severe angioedema can be life threatening due to respiratory obstruction, this is extremely rare in a dermatological context.

Causes of urticaria

Possible allergens, including drugs, should be determined.

Acute and chronic urticaria:

- Autoimmune due to production of antibodies that cross-link the IgE receptor on mast cells
- Allergens (in foods, inhalants and injections)
- Drugs
- Contact, e.g. animal saliva, latex
- Physical, e.g. heat, cold, pressure, sun, water
- Infection, e.g. viral hepatitis, infectious mononucleosis, HIV seroconversion
- Other conditions, e.g. systemic lupus erythematosus (SLE), pregnancy, intestinal parasites
- Idiopathic

Urticarial vasculitis

- Hepatitis B

- SLE
- Idiopathic

Pathogenesis

There is an autoimmune pathogenesis for the most common form of disease



The symptoms and signs of urticaria are due largely to mast cell degranulation with release of histamine and a variety of other vasoactive mediators



More than histamine is clearly involved, as although potent histamine blockers frequently improve the itch of urticaria and the number of weals, they do not abolish them.



Chronic idiopathic urticaria, which is defined by the presence of urticarial episodes for more than 6 weeks



Self-reacting antibodies appear to cause cross-linking of the surface IgE receptor on mast cells with subsequent cellular degranulation.

Clinical history

It is useful to ask to differentiate acute and chronic :

- How long does the individual lesion last? < 24 hours (urticaria)
- > 24 hours (urticarial vasculitis)
- How long has the condition been present? < 6 weeks (acute urticaria)
- > 6 weeks (chronic urticaria)

Distinguishing Urticaria from Urticarial Vasculitis :

1. Knowing the length of time that an individual weal lasts may be of some use in distinguishing urticaria from urticarial vasculitis.
2. Urticarial vasculitis is much less common than urticaria, and many patients are unable to distinguish the development of new weals and disappearance of old ones, from individual weals which persist for more than 1 day.

3. This may be clarified by drawing around a weal with a pen and examining the patient 24 hours later.
4. The physical urticarias can be identified by asking appropriate questions and subsequent challenge testing.

Family history

5. A family history must be sought in cases of angioedema.

Examination

6. Examination may reveal nothing as this is a transient eruption, or may uncover the classical weals which vary from papules to large extensive plaques.

Investigations

- *Full blood count* -including eosinophil count in case of underlying parasites
- *Erythrocyte sedimentation rate (ESR)*: may be elevated in cases of vasculitis
- *Urea and Electrolytes, Thyroid and Liver function tests*: may reveal an underlying systemic disorder
- *Total IgE and specific IgE to possible allergens*, e.g shellfish or peanuts
- *Antinuclear factor*: may be positive in chronic urticaria or urticarial vasculitis
- *CH50*: indicates the level of complement activation; C3 and C4 levels may reveal complement consumption
- *C₁ esterase inhibitor*: may be quantitatively reduced or more rarely functionally deficient, as in hereditary angioedema
- *Skin biopsy*: helpful if urticarial vasculitis is suspected
- *Appropriate challenge test*: to confirm physical urticarias. Frequently no cause can be found for acute episodes, whereas in chronic urticaria an autoimmune pathogenesis will account for the majority of cases.

Psoriasis

1. Psoriasis is a non-infectious, chronic inflammatory disease of the skin, characterised by well-defined erythematous plaques with

silvery scale, with a predilection for the extensor surfaces and scalp, and a chronic fluctuating course.

2. The prevalence is approximately 2% in European populations but may be lower in African and some Asian populations, e.g. Japanese.
3. Psoriasis may start at any age but is unusual before 5 years.
4. There are two epidemiological patterns of psoriasis, sometimes referred to as type 1 and type 2.
5. The first has an onset in the teenage and early adult years, often with a family history of psoriasis and an increased prevalence of the HLA group Cw6.
6. The onset of the second is in the fifties or sixties, when a family history is less common and HLA Cw6 is not so prominent.

Aetiology

1. Psoriasis is genetically complex and a large number of genes are thought to be important in its pathogenesis.
2. There is a large familial component.
3. Formal estimates from twin studies suggest a heritability of around 80%, and in monozygotes perhaps one-third will be concordant for psoriasis.

Factors causing exacerbations of psoriasis

Trauma : Lesions appear in areas of skin damage such as scratches or surgical wounds (Kobner phenomenon) when the condition is erupting

Infection : β -haemolytic streptococcal throat infections often precede guttate psoriasis

Sunlight : UVR rarely

Drugs :

- Antimalarials, (β -adrenoceptor antagonists (β -blockers) and lithium
- 'Rebound' after systemic corticosteroids or potent local corticosteroids are stopped

Emotion : Anxiety

Pathogenesis

There are two key pathophysiological features in psoriatic plaques



The keratinocytes hyperproliferate with a grossly increased mitotic index and an abnormal pattern of differentiation,



leading to the retention of nuclei in the stratum corneum, not ; normally present as the stratum corneum cells are dead.



There is a large inflammatory cell infiltrate many, it is the first presentation of the disease.



The rash often appears rapidly.



Individual lesions are dropletshaped, seldom over 1 cm in diameter and scaly.



Bouts of guttate psoriasis may clear within a few months but respond well to early treatment with phototherapy



The majority of patients go on to develop plaque psoriasis later in life.

Erythrodermic psoriasis

1. The skin becomes universally red or scaly, or more rarely just red with very little scale.
2. As in other forms of erythroderma, temperature regulation becomes compromised with a danger of either hypothermia or hyperthermia.

Pustular psoriasis

1. There are two varieties of pustular psoriasis.

Generalised form

2. The generalised form is rare but serious.
3. The onset is usually sudden, with large numbers of small sterile pustules erupting on a red base.
4. The patient may rapidly become ill with a pyrexia coinciding with the appearance of new

pustules, and will usually require urgent assessment and hospital admission.

A localised form

5. A localised form, which primarily affects the palms and soles, is more common.
6. This is chronic and comprises small sterile pustules which lie on a red base, and resolve to leave brown macules or scaling.

Clinical features

There are several different forms of psoriasis.

Stable plaque psoriasis :

1. Individual lesions are well demarcated and range from a few millimetres to several centimetres in diameter
2. They are red, with a dry silvery-white scale, which may only be obvious after scraping the surface.
3. The elbows, knees and lower back are commonly involved.
4. Other sites of predilection include:
 - Scalp. This is involved in approximately 60% of patients and it is not clear why this is so common.
 - Nails. Involvement is common, with 'thimble pitting', onycholysis (separation of the nail from the nail bed, and subungual hyperkeratosis).
 - Flexures. Psoriasis involving the natal cleft and submammary and axillary folds is not scaly but red, shiny and symmetrical.
 - Palms. Individual plaques may be poorly demarcated and barely erythematous, making this type of psoriasis difficult to differentiate from eczema of the palms.
 - Joint pain (psoriatic arthritis) without any visible skin findings
 - Conjunctivitis or blepharitis

Auspitz's sign

- It is the appearance of punctate bleeding spots when psoriasis scales are scraped off, named after Heinrich Auspitz.
- It is due to thinning of the epidermal layer overlying the tips of the dermal papillae and

blood vessels within the papillae are dilated and tortuous, which bleed readily when the scale is removed.

Guttate psoriasis

This is most commonly seen in children and adolescents, and may follow a streptococcal sore throat.

Lichen Planus

- It is a cell-mediated immune response of unknown origin.
- It may be found with other diseases of altered immunity, such as ulcerative colitis, alopecia areata, vitiligo, dermatomyositis, morphea, lichen sclerosis, and myasthenia gravis.
- A rare type of lichen planus, familial bullous lichen planus, could be gene-related. It may be triggered by diuretics and antimalarials, metal fillings (causing oral lichen planus), stress, and infection.

Clinical presentation

The following may be noted in the patient history:

- Lesions initially developing on flexural surfaces of the limbs, with a generalized eruption developing after a week or more and maximal spreading within 2-16 weeks
- Pruritus of varying severity, depending on the type of lesion and the extent of involvement
- Oral lesions that may be asymptomatic, burning, or even painful
- In cutaneous disease, lesions typically resolve within 6 months (>50%) to 18 months (85%)
- chronic disease is more likely oral lichen planus or with large, annular, hypertrophic lesions and mucous membrane involvement

In addition to the widespread cutaneous eruption, lichen planus can involve the following structures:

- Mucous membranes
- Genitalia
- Nails
- Scalp

The clinical presentation of lichen planus has several variations, as follows:

- Hypertrophic lichen planus
- Atrophic lichen planus
- Erosive/ulcerative lichen planus
- Follicular lichen planus (lichen planopilaris)
- Annular lichen planus
- Linear lichen planus
- Vesicular and bullous lichen planus
- Actinic lichen planus
- Lichen planus pigmentosus
- Lichen planus pemphigoides

Ancillary investigations

- Direct immunofluorescence study reveals globular deposits of immunoglobulin M (IgM) and complement mixed with apoptotic keratinocytes. No imaging studies are necessary.

Distinguishing histopathologic features of lichen planus include the following:

- Hyperkeratotic epidermis with irregular acanthosis and focal thickening in the granular layer
- Degenerative keratinocytes (colloid or Civatte bodies) in the lower epidermis; in addition to apoptotic keratinocytes, colloid bodies are composed of globular deposits of IgM (occasionally immunoglobulin G [IgG] or immunoglobulin A [IgA]) and complement
- Linear or shaggy deposits of fibrin and fibrinogen in the basement membrane zone
- In the upper dermis, a bandlike infiltrate of lymphocytic (primarily helper T) and histiocytic cells with many Langerhans cells

Bullous Pemphigoid

- Bullous pemphigoid is a chronic, inflammatory, subepidermal, blistering disease.
- If untreated, it can persist for months or years, with periods of spontaneous remissions and exacerbations. The disease can be fatal, particularly in patients who are debilitated.

Clinical Presentation- Signs and symptoms

Bullous pemphigoid may present with several

distinct clinical presentations, as follows:

- Generalized bullous form: The most common presentation; tense bullae arise on any part of the skin surface, with a predilection for the flexural areas of the skin
- Vesicular form: Less common than the generalized bullous type; manifests as groups of small, tense blisters, often on an urticarial or erythematous base
- Vegetative form: Very uncommon, with vegetating plaques in intertriginous areas of the skin, such as the axillae, neck, groin, and inframammary areas
- Generalized erythroderma form: This rare presentation can resemble psoriasis, generalized atopic dermatitis, or other skin conditions characterized by an exfoliative erythroderma
- Urticarial form: Some patients with bullous pemphigoid initially present with persistent urticarial lesions that subsequently convert to bullous eruptions; in some patients, urticarial lesions are the sole manifestations of the disease
- Nodular form: This rare form, termed pemphigoid nodularis, has clinical features that resemble prurigo nodularis, with blisters arising on normal-appearing or nodular lesional skin
- Acral form: In childhood-onset bullous pemphigoid associated with vaccination, the bullous lesions predominantly affect the palms, soles, and face
- Infant form: In infants affected by bullous pemphigoid, the blisters tend to occur frequently on the palms, soles, and face, affecting the genital areas rarely; 60% of these infant patients have generalized blisters

Ancillary investigations

To establish a diagnosis of bullous pemphigoid, the following tests should be performed:

- Histopathologic analysis: From the edge of a blister; the histopathologic examination demonstrates a subepidermal blister; the

inflammatory infiltrate is typically polymorphous, with an eosinophil predominance; mast cells and basophils may be prominent early in the disease course

- Direct immunofluorescence (DIF) studies: Performed on normal-appearing, perilesional skin (see the image below), or nonbullous urticarial skin; significant false-negative rate if skin biopsy is taken from skin of the legs
- Direct immunofluorescence study performed on a perilesional skin biopsy specimen from a patient with bullous pemphigoid detects a linear band of immunoglobulin G deposit along the dermoepidermal junction.
- Indirect immunofluorescence (IDIF) studies: Performed on the patient's serum, if the DIF result is positive
- Indirect immunofluorescence study performed on salt-split normal human skin substrate with the serum from a patient with bullous pemphigoid detects immunoglobulin G class circulating autoantibodies that bind to the epidermal (roof) side of the skin basement membrane.
- DIF tests in patients with bullous pemphigoid usually demonstrate immunoglobulin G (IgG) and complement C3 deposition in a linear band at the dermal-epidermal junction, with IgG in salt-split skin found on the blister roof (epidermal side of split skin).
- IDIF studies document the presence of circulating IgG autoantibodies in the patient's serum that target the skin basement membrane component. Seventy percent of patients with bullous pemphigoid have circulating autoantibodies that bind to split skin.

Pemphigus

1. Pemphigus is an uncommon autoimmune disorder that can present with mucosal symptoms only, usually in the mouth.
2. It is usually due to IgG1 and IgG4 autoantibodies, directed against desmogleins-1 and 3 and causing intra-epidermal blistering, but it can also be drug-induced (e.g. by penicillamine

or captopril) and is rarely associated with an underlying malignancy ('para-neoplastic pemphigus').

3. In pemphigus foliaceus, a very superficial form, the autoantibodies are directed solely against desmoglein-1, therefore affecting only the most superficial epidermis..

Clinical features and diagnosis

1. Flaccid blisters occur on the skin, usually on the upper trunk and back.
2. The blisters are superficial; they are fragile and may look like erosions.
3. The Nikolsky sign is positive.
 - It is useful in differentiating between the diagnosis of pemphigus vulgaris or mucous membrane pemphigoid (where the sign is present) and bullous pemphigoid (where it is absent).
 - The Nikolsky sign is dislodgement of intact superficial epidermis by a shearing force, indicating a plane of cleavage in the skin.
4. Mucosal involvement can be severe, increasing the morbidity and risk of mortality.

Investigations

5. Pemphigus can be associated with other autoimmune disorders and full assessment for these is mandatory.
6. In the acute situation, histology from a Tzanck smear (taken from the base of the blister with a scalpel) reveals acantholysis, i.e. separation of the epidermal cells.
7. Disease activity can be monitored by measuring the level of circulating autoantibody in serum by indirect immunofluorescence.
8. Investigations should also include full blood count, ESR, urea and electrolytes, liver function tests, chest X-ray and other imaging, particularly if paraneoplastic pemphigus is suspected.

Mycotic skin diseases

- Fungal infections have been known since the 1000 BC., when Ayurveda Samhitas wrote about Kustha.

- Superficial fungal infections have been known since the 5th century BC, when Hippocrates wrote about thrush in children.
- It took hundreds of years before the first scientific proof of infection was made by Agostino Bassi in 1835, who showed that the muscardine disease of the silkworm was caused by a fungus.
- In the following years Audouin from France suggested that some human diseases were caused by the same types of plant parasites (fungi). By the end of the 19th century important microbiological methods, such as obtaining pure cultures of the dermatophytes *Trichophyton* and *Achorion schoenleinii*, were introduced.
- A morphological classification was not established until 1910, when the famous mycologist R. Sabouraud published "*Les teignes*", a monograph based on the standardization of test media and studies on clinical features of skin and hair infections and morphology in cultures. At the beginning of the 20th century different nomenclatural systems were suggested, based on clinical presentation and culture characteristics.

Etiological agents

Anamorph

- *Epidermophyton*, *Microsporum*, and *Trichophyton* are three anamorphic (asexual or imperfect) genera of the anamorphic class Hyphomycetes of the Deuteromycota that comprise the genus Dermatophytes.
- The genera and descriptions of each are listed below-

<i>Trichophyton</i>	: 24 species
<i>Microsporum</i>	: 16 species
<i>Epidermophyton</i>	: 2 species

Trichophyton spp.

Trichophyton tonsurans is the type species. Microconidia are often more numerous than macroconidia and can be pyriform, globose, clavate,

stalked, sessile, and borne individually along the edges of the hyphae or in clusters that resemble grapes.

Microsporum spp

Microsporumma fouinii is the type species.

Emmons first classified the macroconidia as spindle- or fusiform-shaped, but the discovery of additional species, such as *Microsporum naman*, increased the range from obovate (egg-shaped) to cylindrofusi form

Epidermophyton spp.

Epidermophyton floccosum, the sole species of its kind, is anthropophilic, unable to dwell freely in soil, and only sometimes seen to parasitize lesser animals

Teleomorphs

Only a few species of dermatophytes, primarily the geophilic and zoophilic *Trichophyton* and *Microsporum* species, are capable of sexual reproduction and the production of ascospores.

Ecology

Based on the ecological niches in which they normally reside, dermatophytes have been divided into geophilic, zoophilic, and anthropophilic species. (Spiliopoulou A et al., 2015).

Geophilic

These fungi may infect people and animals and are commonly isolated from the soil where they are linked to keratinous material that is degrading.

Anthropophilic

These dermatophytes are mostly connected to humans and seldom infect other species. Direct contact or indirect transmission by fomites is the two ways these groupings of the organism are spread.

Zoophilic

Zoophilic dermatophytes frequently infect animals and sporadically humans.

This fungus may infect individuals either directly through contact with an infected animal or indirectly through contact with spores.

Ecological classification of dermatophytes.

Geophilic	Anthrophilic	Zoophilic
<i>Trichophyton ajelloi</i> <i>Trichophyton terrestre</i>	<i>dermatophyton floccosum</i> <i>Microspor- um audouinii</i> <i>Microsporum ferrugineum</i> <i>Trichophyton concentricum</i> <i>Trichop- hyton kanei</i> <i>Trichophyton megninii</i> <i>Trichophyton mentagrophytes</i> <i>Tricho- phyton raubitschekii</i> <i>Trichophyton</i> <i>rubrum</i> <i>Trichophyton schoenleinii</i> <i>Trichophyton soudanense</i> <i>Trichophyton</i> <i>tonsurans</i>	<i>Microsporum canis</i> <i>Microsporu- mequinum</i> <i>Microsporumpersi- color</i> <i>Microsporumnannum</i> <i>Trichophyton mentagrophyte</i> <i>Trichophyton simii</i> <i>Trichophyton</i> <i>equinum</i> <i>Trichophyton</i> <i>verrucosum</i>

Epidemiology

WHO estimates that dermatophytes are the major organism responsible for the millions of individuals infected by superficial fungal infections globally, with an estimated lifetime risk of 20–25%. (World Health Organization; 2005).

Although tinea infections are widespread worldwide, they are more frequent in tropical nations like India. Dermatophytosis varies from place to place, person to person, and is caused by a variety of factors, including socioeconomic situations, climate, and population density. Dermatophytosis is becoming more common due to a number of causes, including the tropical temperature, urbanisation, overcrowding, shared housing, wearing tightly-fitting clothing, wearing occlusive footwear, participating in sports, and public bathing.

While tinea capitis is more frequently found in adolescents and children, tinea cruris and tinea pedis are the most prevalent illnesses in adult males.

Clinical Manifestation

Tinea capitis

Tinea capitis, the dermatophyte infection commonly affects scalp and hair, caused by members of the genera *Microsporum* and *Trichophyton*, most commonly occurs in children.

It may range from mild, almost subclinical, with slight erythema and a few patchy areas of scaling with dull gray hair stumps to a highly inflammatory reaction with folliculitis, kerion formation, and

extensive areas of scarring and alopecia, sometimes accompanied by fever, malaise, and regional lymphadenopathy. Ectothrix, the infection of the hair, appears as a sheath of arthroconidia formed on the outside of the hair shaft.

Pathogens associated with clinical types of Tinea Capitis

Inflammatory	<i>Microsporumaudouinii</i> , <i>Microsporumcanis</i> , <i>Microsporumgypseum</i> , <i>Microsporumnannum</i> <i>Trichophyton interdigitale</i> , <i>Trichophyton schoenleinii</i> , <i>Trichophyton tonsurans</i> , <i>Trichophyton verrucosum</i>
Black dot	<i>T. tonsurans</i> , <i>T. violaceum</i>
Non-inflamm- atory	<i>M. audouinii</i> , <i>M. canis</i> <i>M. ferrugineum</i> , <i>T. tonsurans</i>
Favus	<i>T. schoenleinii</i> , <i>T. violaceum</i> , <i>T. mentagrophytes</i>

Tinea corporis

Tinea corporis (i.e. body ringworm) occurs on the trunk and extremities. The infection may range from mild to severe, commonly appearing as annular erythematous scaly patches with a raised border and central clearing. Varying degrees of itching is present in the lesion.

T. rubrum and *E. floccosum* are anthropophilic organisms that cause tinea corporis and infect the skin but not the hair, show little inflammation and very little scaling at the edge lead to the chronic lesion. Other dermatophytes also found in tinea capitis, such as *M. audouinii*, *T. schoenleinii*, *T.*

tonsurans and *T. violaceum* can also cause tinea corporis.

Tinea barbae

Tinea barbae defines the infection of the skin and coarse hairs of the beard and mustache area, which occurs in adult men and hirsute women

Tinea faciei

Tinea faciei involves the infection of the nonbearded areas of the face. Affected patients may have burning and itching, gets aggravated by exposure to sunlight.

Tinea cruris

Tinea cruris, or "jock itch," is a dermatophyte infection of the groin (particularly the proximal medial thighs), perineal and perianal areas, may extend to the buttocks or abdomen, but usually does not involve the scrotum

Tinea pedis

Tinea pedis (Athlete's foot) involves infection of the feet, especially the soles and toe webs. There are three common presentations of tinea pedis. The most common is the interdigital type, which is characterized by itching, maceration, burning sensation, fissuring and peeling in between the toes.

Tinea manuum

Tinea manuum is a dermatophytosis of one or both hands. It is most commonly caused by *Trichophyton* spp. Palms are typically hyperkeratotic and very dry; fingernails may be involved.

Onychomycosis

Onychomycosis describes fungal infection of the nail caused by a fungus such as dermatophytes, nondermatophyte molds or yeasts. Tinea unguium refers strictly to dermatophyte infection of the nail, accounts for approximately 90% of cases

- 1) Distal and Lateral Subungual Onychomycosis (DLSO): it is the most common type (90%), usually caused by *T. rubrum*. Discoloration, subungual hyperkeratosis and distal onycholysis start at the hyponychium and spreads proximally.

- 2) Proximal Subungual Onychomycosis (PSO): the dermatophytes invade the nail unit under the proximal nail fold and spread distally. It is usually caused by *T. rubrum* and is usually associated with immunosuppressed conditions, e.g. HIV infection.
- 3) Superficial White Onychomycosis (SWO): the fungi, mainly 7 mentagrophytes, directly invade the superficial layers of the nail plate but do not penetrate it leading to a white, crumbly nail surface.
- 4) Total Dystrophic Onychomycosis: It is characterized by thickened, opaque, discolored, broken and dystrophic nails. The nail plate may be separated from the nail bed. Toenails are affected more often than fingernails, and individual nails are sometimes spared.

Tinea imbricata

Tinea imbricata is the chronic infectious manifestation of tinea corporis, characterized by concentric rings of overlapping scales scattered throughout the body.

Tinea incognito

Tinea incognito is mycosis of the skin caused by fungi such as *T. rubrum*, *T. mentagrophytes*, *E. floccosum*, *M. canis*, *M. gypseum* and *M. jansel* seen in immunocompromised. Symptoms are skin atrophy, telangiectasia and florid growth (combinations of papules, pustules or nodules) on feet, ankle, legs or groin.

Majocchi's Granuloma:

Majocchi's granuloma occurs when dermatophytes invade the dermis and subcutaneous tissues, usually via disrupted hair follicles, as a result of trauma or shaving, largely on the extremities, occasionally on the scalp or face.

Id Reaction:

The id reaction is an acute, local or more generalized, papulovesicular eruption that usually occurs at a site remote from the site of dermatophyte infection. It occurs due to sensitization to circulating antigens and not from direct infection

by dermatophytes or other fungal, bacterial, viral or parasitic infections, The id reaction should not be mistaken for a drug reaction.

Investigations

Dermatophytosis is usually diagnosed by a combination of direct microscopic examination, culture and Wood's lamp examination. Biopsy (histopathology) tends to be used mainly when the presentation is unusual.

Wood's Lamp Examination

Wood's lamp is a device that is useful in diagnosis and management of superficial cutaneous fungal infections. Wood's glass consists of barium silicate containing about 9% nickel oxide. It transmits long-wave ultraviolet light with peak of 365 nm that shows characteristic fluorescence produced by some microbial agents. This fluorescence has been used to demonstrate hair infection mainly by microsporum species and one of the trichophyton species. It can also be helpful in detecting some species of dermatophytes, such as *M. canis* and *T. quinckeanum*. Not all strains of these organisms exhibit fluorescence,

The chemical substance responsible for positive fluorescence is pteridine. Therefore, other materials containing this substance may also give false-positive fluorescence. The commonest source of error is bluish or purplish fluorescence produced by ointments containing petroleum jelly, scales, serum exudates, lint and dried soap.

Fluorescence produced by different microorganisms under Wood's lamp

Microorganisms	Fluorescence Color
<i>Microsporum audouinii</i>	Bright-green
<i>Microsporum canis</i>	Bright-green
<i>Microsporum ferrugineum</i>	Blue-green
<i>Microsporum distortum</i>	Blue-green
<i>Microsporum gypseum</i>	Dull-yellow
<i>Trichophyton schoenleinii</i>	Dull-green

Laboratory Diagnosis

The diagnosis of dermatophytosis is based on combination of clinical observation and laboratory

investigations depend on demonstration of causative pathogen in skin-scraping by microscopy, culture, serological and molecular test

Collection and Transport of Specimens

Dermatophytes shows radial growth and the centers of infected skin patches consist of the older and the poorly viable material. The clinical specimen like skin scales near the advancing edges of the rings, nail clipping or curettings from dystrophic subungual debris and hair stubs are usually examined. Firstly, the lesion is lightly disinfected with alcohol in gauze and then it is scraped from center to edge, crossing the lesion margin by using a sterile scalpel blade or equivalent.

If lesions consist of vesicles or bullae, then top of the vesicle or bulla should be collected as a sample. In tinea capitis and tinea barbae, basal root portion of the hair is best for direct microscopy and culture. From nails, close clippings of the whole nail end are taken from an appropriate site depending on the type of nail infection.

Direct Examination

Despite false negatives in 5 to 15% of cases in ordinary practice, the microscopic examination of KOH wet mounts of keratinous material is very simple and reliable screening technique. Scrapings may be mounted for direct examination in 10% KOH for skin scrapings and 20-40% KOH for hair and nail samples.

In the KOH wet mount, fungus is seen as branching hyaline mycelia, which frequently show arthrospores production particularly in the infected skin appendages like hair. Ectothrix and endothrix types of infections are distinguished in wet mount and arrangement of arthrospores can be made out. The demonstration of fungus in nails may be difficult and may be possible only after keeping clippings in KOH for overnight.

Fungal Culture

The clinical specimens should be inoculated on fungal culture media irrespective of the findings of direct examination. Culture is essential and valuable adjunct to direct microscopy.

Dermatophytes can also grow easily on Sabouraud's Dextrose Agar with antibiotics or cycloheximide and the media should be incubated at three temperatures i.e. 25°C, 30°C and 37°C. The growth is relatively slow and usually takes about 10 days to three weeks time.

The typical isolates of common dermatophytes that can be identified directly from primary isolation media like Sabouraud's Dextrose Agar and Potato Glucose or Potato Flake Agar on the basis of identification characteristics like colony pigmentation, texture, and growth rate and distinctive morphological structures, such as microconidia, macroconidia, spirals, pectinate branches, pedicels, and nodular organs.

Culture is a precious accessory of microscopy which is crucial in all infections for treatment with systemic prescription.

BCP-milk solids-glucose agar is used to differentiate dermatophytes such as *T. rubrum*, *T. mentagrophytes*, *T. soudanense*, *T. megninti*, *M. persicolor* and *M. equinum*, on the basis of catabolite domination by glucose and divergence of releasing ammonium ions from casein.

Littman ox gall agar as restrictive media is preferred to diagnose a non dermatophyte infection.

Casamino acids-erythritol-albumin medium was firstly used by Fischer and Kane. This medium contains balanced egg albumin, which reduces yeasts such as *C. albicans* and is an extremely usable medium for isolating dermatophytes from heavily contaminated samples.

Bromocresol purple (BCP) casein-yeast extract agar is designed for the rapid recognition of microcolonies of *T. verrucosum* though all dermatophytes are able to grow on this medium.

Biochemical and Physiological Tests-

Urease Activity

Some dermatophytes are able to hydrolyse high concentrations of urea. This test is useful to differentiate *T. rubrum* (urease negative) from *T. mentagrophytes* var. *interdigitale* (urease positive).

In vitro Hair Perforation

Some dermatophytes are able to penetrate and

invade hair shaft by producing specialized perforating organs in vitro, while other species attack the hair by simple peripheral erosions.

Mating Test

Dermatophytes are heterothallic species. In order to obtain a sexual reproduction, it is necessary to confront strains from the same species with complementary (+ and -) isolates. To observe the teleomorph (*Arthroderma* genus) state of *Microsporum* or *Trichophyton* species, reference strains with complementary polarity are required. Takashio's or De Vroey's (*Guiauteabyssinica* seeds) media, that are poor culture media stimulating the sexual conidiation of fungi, can be used.

Immunodiagnosis

The skin test with dermatophytic antigen, trichophytin and serological tests are important for diagnosis of dermatophytosis. The trichophytin is crude extract from dermatophytes, producing positive delayed-type hypersensitivity, tuberculin-like response in most of the adults.

Molecular diagnosis of dermatophytosis

Nucleic acid-based molecular methods are increasingly being employed for rapid and specific identification of the fungi as well as for the detection of etiological agent directly from the clinical specimen in the clinical microbiological laboratory.

Leprosy

1. Leprosy (Hansen's disease) is a chronic granulomatous disease affecting skin and nerves, caused by *Mycobacterium leprae*, a slow-growing mycobacterium which cannot be cultured in vitro.
2. The clinical manifestations are determined by the degree of the patient's cell-mediated immunity (CMI) towards *M. leprae*.
3. High levels of CMI with elimination of leprosy bacilli produce tuberculoid leprosy, whereas absent CMI results in lepromatous leprosy.
4. The complications of leprosy are due to nerve damage, immunological reactions and bacillary infiltration.

5. Leprosy patients are frequently stigmatised and using the word 'leper' is inappropriate.

Epidemiology and transmission

1. Some 4 million people have leprosy and around 750 000 new cases are detected annually.
2. 70% of the world's leprosy patients live in India, with the disease endemic in Brazil, Indonesia, Mozambique, Madagascar, Tanzania and Nepal.

Spread of infection

3. Untreated lepromatous patients discharge bacilli from the nose.
4. Infection occurs through the nose, followed by haematogenous spread to skin and nerve.
5. The incubation period is 2-5 years for tuberculoid cases and 8-12 years for lepromatous cases.
6. Leprosy incidence peaks at 10-14 years, and is more common in males and in those with close household exposure to leprosy cases.

Pathogenesis

M. leprae has a predilection for infecting Schwann cells and skin macrophages.



In tuberculoid leprosy, effective CMI controls bacillary multiplication ('paucibacillary') but organised epithelioid granulomas are formed.



In lepromatous leprosy, there is abundant bacillary multiplication ('multibacillary'), e.g. in Schwann cells and perineurium.



Between these two extremes is a continuum, varying from patients with moderate CMI (borderline tuberculoid) to patients with little cellular response (borderline lepromatous).



Associated with immunological reactions to the infection occur as the immune response

develops and the antigenic stimulus from the bacilli varies



particularly in borderline patients.



Delayed hypersensitivity reactions produce type 1 (reversal) reactions, while immune complexes contribute to type 2 (erythema nodosum leprosum) reactions.



HIV/leprosy co-infected patients have typical lepromatous and tuberculoid leprosy skin lesions and typical leprosy histology and granuloma formation.



Surprisingly, even with low circulating CD4 counts, tuberculoid leprosy may be observed and there is not an obvious shift to lepromatous leprosy.

Clinical features

Cardinal features of leprosy

- Skin lesions, typically anaesthetic at tuberculoid end of spectrum
- Thickened peripheral nerves
- Acid-fast bacilli on skin smears or biopsy

Clinical characteristics of the polar forms of leprosy

Clinical and tissue specific features	Lepromatous	Tuberculoid
Skin and nerves		
Number and distribution	Widely disseminated	One or a few sites asymmetrical
Skin lesions		
Definition		
Clarity of margin	Poor	Good
Elevation of margin	Never	Common
Colour		
Dark skin	Slight	Marked
hypopigmentation	hypopigmentation	
Light skin	Slight erythema	Coppery or red

Surface	Smooth, shiny	Dry, scaly
Central healing	None	Common
Sweat and hair growth	Impaired late	Impaired early
Loss of sensation	Late	Early and marked
Nerve enlargement and damage	Late	Early and marked
Bacilli (bacterial index)	Many (5 or 6+)	Absent (D)
Natural history	Progressive	Self-healing
Other tissues	Upper respiratory mucosa, eye, testes, bones, muscle	None
Reactions	Immune complexes (type 2)	Cell-mediated (type 1)

Leprosy reactions

- **Type 1 (reversal) reactions** : These occur in 30% of borderline patients (BT, BB or BL) and are delayed hypersensitivity reactions. Skin lesions become erythematous. Peripheral nerves become tender and painful, with sudden loss of nerve function. These reactions may occur spontaneously, after starting treatment and also after completion of multidrug therapy.
- **Type 2 (erythema nodosum leprosum-ENL) reactions** : These are partly due to immune complex deposition and occur in BL and LL patients who produce antibodies and have a high antigen load. They manifest with malaise, fever and crops of small pink nodules on the face and limbs. Iritis and episcleritis are common. Other signs are acute neuritis, lymphadenitis, orchitis, bone pain, dactylitis, arthritis and proteinuria. ENL may continue intermittently for several years.

Reactions in leprosy

	Leprosy reaction	Leprosy reaction type 2 type 1 (reversal) (erythema nodosum leprosum, ENL)
Mechanism	Cell-mediated hypersensitivity	Immune complexes
Clinical	Painful tender	Tender papules and

features	nerves, loss of function Swollen skin lesions New skin lesions lymphadenitis Fever, oedema	nodules; may ulcerate Painful tender nerves, loss of function Iritis, orchitis, myositis,
Management	Prednisolone 40 mg, reducing over 3-6 months'	Moderate: prednisolone 40 mg daily Severe: thalidomide or prednisolone 40-80 mg daily, reducing over 1-6 months; local if eye involved

Borderline cases

1. In borderline tuberculoid (BT) cases, skin lesions are more numerous than in tuberculoid (TT) cases, and there is more severe nerve damage and a risk of type 1 reactions.
2. In borderline leprosy (BB) cases, skin lesions are numerous and vary in size, shape and distribution; annular lesions are characteristic and nerve damage is variable.
3. In borderline lepromatous (BL) cases, there are widespread small macules in the skin and widespread nerve involvement; both type 1 and type 2 reactions occur.
4. Pure neural leprosy (i.e. without skin lesions) occurs principally in India and accounts for 10% of patients.
5. There is asymmetrical involvement of peripheral nerve trunks and no visible skin lesions.
6. On nerve biopsy all types of leprosy have been found.

Investigations

1. The diagnosis is clinical, made by finding a cardinal sign of leprosy and supported by finding acid-fast bacilli in slit-skin smears or typical histology in a skin biopsy.
2. Slit-skin smears are obtained by scraping dermal material on to a glass slide. The smears are then stained for acid-fast bacilli, the number

counted per high-power field and a score derived on a logarithmic scale (0-6): the bacterial index (BI).

3. Smears are useful for confirming the diagnosis and monitoring response to treatment.
4. Neither serology nor PCR testing for *M. leprae* DNA is sensitive or specific enough for diagnosis.

Vitiligo

- Vitiligo is an acquired pigmentary disorder of the skin that is characterized by circumscribed, depigmented macules and patches.
- The condition is frequently associated with disorders of autoimmune origin, with thyroid abnormalities being the most common.

Clinical presentation -Signs and symptoms

Vitiligo lesions are characterized as follows:

- White or depigmented macules and patches
- Usually well demarcated
- Round, oval, or linear in shape
- Borders may be convex
- Range from millimeters to centimeters in size
- Enlarge centrifugally over time at an unpredictable rate
- Initial lesions occur most frequently on the hands, forearms, feet, and face, favoring a perioral and periocular distribution.

Clinical variants

- Trichrome vitiligo
- Marginal inflammatory vitiligo
- Quadrichrome vitiligo
- Koebner phenomenon: Development of vitiligo in sites of specific trauma, such as a cut, burn, or abrasion.

Ancillary investigation

- Although the diagnosis of vitiligo generally is made on the basis of clinical findings, biopsy is occasionally needed for differentiating vitiligo from other hypopigmenting or depigmenting disorders.
- Microscopic examination of involved skin shows a complete absence of melanocytes in

association with a total loss of epidermal pigmentation.

- Superficial perivascular and perifollicular lymphocytic infiltrates may be observed at the margin of vitiginous lesions, consistent with a cell-mediated process destroying melanocytes.

Other documented histologic findings include the following:

- Degenerative changes in keratinocytes and melanocytes in the border lesions and adjacent skin
- Increased numbers of Langerhans cells
- Epidermal vacuolization
- Thickening of the basement membrane
- Loss of pigment and melanocytes in the epidermis is highlighted by Fontana-Masson staining and immunohistochemistry testing.

Cellulitis

- The term cellulitis commonly is used to indicate a nonnecrotizing inflammation of the skin and subcutaneous tissues, usually from acute infection.
- Cellulitis usually follows a breach in the skin, although a portal of entry may not be obvious; the breach may involve microscopic skin changes or invasive qualities of certain bacteria.

Etiology-Clinical presentation -Signs and symptoms

Nonpurulent cellulitis is associated with the four cardinal signs of infection, as follows :

- Erythema
- Pain
- Swelling
- Warmth

Physical examination findings that suggest the most likely pathogen include the following:

- Skin infection without underlying drainage, penetrating trauma, eschar, or abscess is most likely caused by streptococci
- *Staphylococcus aureus*, often community-

acquired MRSA, is the most likely pathogen when these factors are present.

- Violaceous color and bullae suggest more serious or systemic infection with organisms such as *Vibrio vulnificus* or *Streptococcus pneumoniae*

The following findings suggest severe infection:

- Malaise, chills, fever, and toxicity
- Lymphangitic spread (red lines streaking away from the area of infection)
- Circumferential cellulitis
- Pain disproportionate to examination findings

Indications for emergent surgical evaluation are as follows :

- Violaceous bullae
- Cutaneous hemorrhage
- Skin sloughing
- Skin anesthesia
- Rapid progression
- Gas in the tissue
- Hypotension

Ancillary investigations

Generally, no workup is required in uncomplicated cases of cellulitis that meet the following criteria:

- Limited area of involvement
- Minimal pain
- No systemic signs of illness (eg, fever, altered mental status, tachypnea, tachycardia, hypotension)
- No risk factors for serious illness (eg, extremes of age, general debility, immunocompromise)

The Infectious Disease Society of America (IDSA) recommends the following blood tests for patients with skin or soft tissue infection (SSTI) who have signs and symptoms of systemic toxicity:

- Blood cultures
- CBC with differential
- Levels of creatinine, bicarbonate, creatin phosphokinase, and C-reactive protein (CRP)

Blood cultures should also be done in the following circumstances :

- Moderate to severe disease (eg, cellulitis complicating lymphedema)
- Cellulitis of specific anatomic sites (eg, facial and especially ocular areas)
- Patients with a history of contact with potentially contaminated water
- Patients with malignancy who are receiving chemotherapy
- Neutropenia or severe cell-mediated immunodeficiency
- Animal bites

Other tests to consider are as follows:

- Mycologic investigations are advisable if recurrent episodes of cellulitis are suspected to be secondary to tinea pedis or onychomycosis
- Creatinine levels help assess baseline renal function and guide antimicrobial dosing

Imaging studies

- Ultrasonography may play a role in the detection of occult abscess and direction of care
- Ultrasonographic-guided aspiration of pus can shorten hospital stay and fever duration in children with cellulitis
- If necrotizing fasciitis is a concern, CT imaging typically is used in stable patients; MRI can be performed, but MRI typically takes much longer than CT scanning
- Strong clinical suspicion of necrotizing fasciitis should prompt surgical consultation without delay for imaging

Aspiration, Dissection, and Biopsy

- Needle aspiration should be performed only in selected patients or in unusual cases, such as in cases of cellulitis with bullae or in patients who have diabetes, are immunocompromised, are neutropenic, are not responding to empiric therapy, or have a history of animal bites or immersion injury .
- Aspiration or punch biopsy of the inflamed area may have a culture yield of 2-40% and is of limited clinical value in most cases

- Gram stain of aspiration or biopsy specimens has a low yield and is unnecessary in most cases, unless purulent material is draining or bullae or abscess is present; however, Gram stain and culture after incision and drainage of an abscess yields positive results in more than 90% of cases
- Dissection of the underlying fascia to assess for necrotizing fasciitis may be determined by surgical consultation or indicated following initial evaluation and imaging studies
- Skin biopsy is not routine but may be performed in an attempt to rule out a noninfectious entity

Hospital admission

The IDSA recommends considering inpatient admission in patients with hypotension and/or the following laboratory findings:

- Elevated creatinine level
- Elevated creatine phosphokinase level (2-3 times the upper limit of normal)
- CRP level >13 mg/L (123.8 mmol/L)
- Low serum bicarbonate level
- Marked left shift on the CBC with differential

MCQs

1. Leonine facies seen in

- a. Tuberculosis b. Leprosy
c. Amyloid d. AIDS

2. In which skin disorder do T-cells induce a sudden increase in cytokines leading to hyperplasia of skin cells resulting in raised skin plaques?
 - a. dermatitis
 - b. eczema
 - c. psoriasis
 - d. impetigo
3. Which among the following is not correct regarding severe cellulitis
 - a. Elevated creatinine level
 - b. Elevated creatine phosphokinase level (2-3 times the upper limit of normal)
 - c. CRP level >13 mg/L (123.8 mmol/L)
 - d. High serum bicarbonate level
4. Which of the following is the most common site of cellulitis?
 - a. Face
 - b. Hand
 - c. Leg
 - d. Arm
5. Which of the following pathogens is most likely responsible for a skin infection with underlying drainage, penetrating trauma, eschar, or abscess?
 - a. Staphylococcus aureus
 - b. Vibrio vulnificus
 - c. Streptococcus pneumoniae
 - d. Pseudomonas aeruginosa

Answer

1. b	2. c	3. d
4. c	5. a	